

UNIT-5

Correlations and Regressions(memory sheet)

1. Correlation Definition Correlation measures the degree and direction of relationship between two variables.	4. Least Squares (Straight Line) Equation $y = a + bx$
Karl Pearson's Coefficient (r) Formula $r = \Sigma XY / \sqrt{(\Sigma X^2 \times \Sigma Y^2)}$	Normal Equations $\Sigma y = na + b\Sigma x$ $\Sigma xy = a\Sigma x + b\Sigma x^2$
Range $-1 \leq r \leq 1$ $r = +1 \rightarrow$ perfect positive $r = -1 \rightarrow$ perfect negative $r = 0 \rightarrow$ no correlation	Table Needed $ x y x^2 xy $ 5. Straight Line (ax + b form) Equation $y = ax + b$
Important Properties - Unit free - Symmetric: $r(x,y) = r(y,x)$ - Independent of origin & scale - Measures only linear relationship - Affected by extreme values	Normal Equations $\Sigma y = a\Sigma x + nb$ $\Sigma xy = a\Sigma x^2 + b\Sigma x$ 6. Exponential Curve
2. Rank Correlation (Spearman) Formula $\rho = 1 - [6\Sigma d^2 / n(n^2 - 1)]$	Curve equation $y = ae^{bx}$
With Ties (Important) Add correction: $\rho = 1 - [6(\Sigma d^2 + \text{correction}) / n(n^2 - 1)]$	Transformation $\log y = \log a + bx$ Let: $Y = \log y$ $A = \log a$ 👉 $Y = A + bx$
3. Regression (VERY IMPORTANT)	
Regression Lines $y - \bar{y} = b_{yx} (x - \bar{x})$ $x - \bar{x} = b_{xy} (y - \bar{y})$	Use same formulas $\Sigma Y = nA + b\Sigma x$ $\Sigma xY = A\Sigma x + b\Sigma x^2$
Regression Coefficients $b_{yx} = \Sigma XY / \Sigma X^2$ $b_{xy} = \Sigma XY / \Sigma Y^2$	7. Quadratic Curve Equation $y = a + bx + cx^2$
Relation with r $r = \sqrt{(b_{yx} \times b_{xy})}$	Normal Equations $\Sigma y = na + b\Sigma x + c\Sigma x^2$ $\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$ $\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$
Key Concept Both regression lines intersect at $(\bar{x}, \bar{y})$	Table Needed $ x y x^2 x^3 x^4 xy x^2 y $